Causal Sets by Dynamical Densification: Continuum Limit, Order-Only Rules, and Phase Interference

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Abstract

We study a growth model for causal sets in which every new element is inserted into the interior of the existing causal set, so that the density of elements increases dynamically. The insertion rule is Lorentz invariant: a child is placed at a small, Poisson-distributed proper-time separation from a uniformly chosen parent, with a rapidity drawn from the Haar measure $\sinh^2\eta\, d\eta$, optionally damped by a Maxwell–Jüttner factor $e^{-\alpha\cosh\eta}$ whose rest frame is read off the state. We use the model to address four open problems of causal set theory.

Continuum limit. The empirical measure of the growing set converges almost surely for every $\alpha \geq 0$, and every finite induced-subposet density converges, so the process has a poset limit: a uniform sprinkling of a conformally flat spacetime. For $\alpha = 0$ the limit is explicit, $a^4 \propto t^2 + (T-t)^2$.

Mesoscopic manifoldlikeness. Exact second-moment recursions, an eigenvalue criterion and martingale concentration give high-probability versions of Madsen's conditions F2 (counting equals volume, uniformly over all mesoscopic causal diamonds) and F3 (longest chain equals proper time) for the conformal metric. Density clumping, a Pólya-urn effect, is confined to wavelengths above 8.0–9.1 damping lengths.

Lorentz invariance versus locality. Frame independence holds at the 1–5% resolution of intrinsic tests. Lorentz violation enters only through the clumping excess and is negligible in the physical regime.

Rogue posets and quantum dynamics. Order-only insertion rules fail in three distinct ways, all traced to the light-cone nonlocality of order-space neighbours, whose symmetric difference grows as $N^{1/2}$ in 2D and $N^{0.8}$ in 4D. Attaching the phase $e^{i\beta S_{\mathrm{BD}}}$ to such histories, we prove a defect-gas theorem: for M independent local defect sites with K alternatives and uniformly distributed relative phases, the defect amplitude grows as $e^{M\, m(1+x_1+\cdots+x_K)}$, where m is the Mahler measure. The rate is zero for $K = 1$ and positive for $K \geq 2$, so multi-type local defects are not suppressed by interference. Numerical checks support the uniform-phase hypothesis at generic β, additivity, the overlap correction and the absence of double counting. The key analytic steps are machine-checked in Lean 4 with Mathlib.

I Introduction

A causal set is a locally finite partial order whose elements are spacetime atoms and whose order is the causal relation [1, 2]. The programme rests on a kinematic fact and a dynamical hope. The fact is that a Poisson sprinkling of a Lorentzian manifold produces a causal set that is statistically Lorentz invariant [3]. The hope is that some dynamics will make such manifoldlike causal sets dominate. Between the fact and the hope stand four well-known problems [2, 4].

1. **Rogue posets.** Almost all partial orders on N elements are three-layer Kleitman–Rothschild (KR) orders, about $2^{N^2/4}$ of them [5]. Any dynamics must overcome this entropy, and the entropy of every other non-manifold class.
2. **Quantum dynamics.** The natural candidate is a sum over causal sets weighted by $e^{i\beta S_{\mathrm{BD}}}$, with S_{BD} the Benincasa–Dowker action [6, 7] and Sorkin's quantum measure as the interpretive frame [8]. Layered orders, KR orders included, are known to be strongly suppressed by the phase [9–11]. What survives is not known [4].
3. **Lorentz invariance versus locality.** Sprinkled causal sets are Lorentz invariant because their links reach arbitrarily far along the light cone. Rules that are local in any frame break Lorentz invariance; rules that are Lorentz invariant are nonlocal [3, 12].
4. **Continuum limit.** It is not settled in what sense a growing causal set converges to a manifold. For classical sequential growth [13] the order-theoretic limits are understood [14]. For growth that inserts elements into the past, even the choice of observables is contested [15].

This paper studies one model against all four problems. Its defining choice is **dynamical densification**: each new element is inserted into the interior of the existing set, at a small proper-time separation from a randomly chosen parent, with uniformly random rapidity. Classical sequential growth instead adds elements only to the future boundary, so that birth order is a time order. Densification has three structural advantages. A superposition of independent Poisson sprinklings is again a Poisson sprinkling, so a rule that inserts uniformly by volume keeps every intermediate stage a Lorentz-invariant sprinkling. The continuum limit is then simply the limit of increasing density. And the growth index becomes a resolution scale rather than a time, which removes the requirement that the past be settled. The price is that inserting an element correctly requires knowing its relation to elements far away along the light cone. Much of what follows measures that price.

We call the coordinate version of the rule the *replication model*. A child y of parent p is placed at proper time τ with $\tau^4 \propto E$, $E \sim \mathrm{Exp}(1)$ in units of one element per discreteness volume, and at rapidity η drawn from the Haar measure $\sinh^2\eta\, d\eta\, d\Omega$ of the Lorentz group, optionally damped by $e^{-\alpha\cosh\eta}$ in a rest frame estimated from the state. The parameter α is the ratio of the discreteness length to a damping length $L_H = \ell/\alpha$, which we identify physically with the Hubble radius, so $\alpha \sim 10^{-61}$ today.

A Summary of results

Problem	Result	Status
Continuum limit	Empirical measure converges a.s. for all $\alpha \geq 0$; all induced-subposet densities converge; the poset limit is a uniform sprinkling of $(M, a^2\eta)$; explicit for $\alpha = 0$ (Theorems 17–20, Sec. V)	Proof; abstract layer in Lean
Mesoscopic manifoldlikeness	High-probability F2 uniformly over all diamonds and F3 with exponential tails, for the conformal metric; clumping only above 8.0–9.1 L_H (Propositions 1–3, 7; Theorems 4D, 5D, 8–10, 13, 16; Lemmas 6D, 11, 12; Sec. IV)	Proof plus numerics
Lorentz versus locality	Frame independence at 1–5%; violation only through the clumping excess, $\propto (H/M_{\mathrm{Pl}})^2$ in the physical regime (Sec. III)	Numerics plus estimates

Problem	Result	Status
Rogue posets	Coordinate-model output passes all intrinsic tests; order-only insertion rules fail in three classified ways, traced to nearest-neighbour symmetric differences $\propto N^{1/2}$ (2D) and $\propto N^{0.8}$ (4D) (Sec. VI)	Numerics
Quantum dynamics	Defect-gas theorem: rate = Mahler measure, 0 for $K = 1$ and > 0 for $K \geq 2$; KR threshold reproduced; interactions raise the β-averaged rate (Theorem 21, Sec. VII)	Proof; core steps in Lean; numerics

The overall picture is a division of labour. The replication model supplies a kinematics that is Lorentz invariant at the discreteness scale and converges to a conformally flat continuum. The intrinsic test battery and the poset-limit theorems verify that output using the order alone. The quantum phase cannot replace a well-behaved classical measure: it suppresses defects whose action shifts are concentrated enough to phase-lock, as for layered and KR orders, but not the multi-type local defects that order-only growth rules generate.

Three limitations should be stated at the outset. The replication rule places children in background coordinates, so it is not a purely order-theoretic dynamics; Section VI explains why we believe no simple order-only rule can replace it. The large-scale conformal factor is fixed by boundary conditions, not by the dynamics (Sec. II C). And Theorem 21 assumes uniformly distributed relative phases, a hypothesis we test numerically rather than prove.

II The replication model

The replication model grows a causal set inside a fixed region by repeatedly choosing a parent uniformly among the existing elements and inserting one child near it. "Near" is defined invariantly: the child lies at a small proper time from the parent, in a uniformly random direction on the hyperboloid. All results in this section are numerical and set up the theorems of Secs. IV and V.

A The 1+1-dimensional rule

In two dimensions write the displacement in light-cone coordinates. The child of p is placed at $(\Delta u, \Delta v) = (\sqrt{s}\, e^{\eta}, \sqrt{s}\, e^{-\eta})$, where s is the invariant interval volume, chosen so that the parent–child diamond contains on average $c\,\mathrm{Exp}(1)$ elements, and η is uniform on the allowed range (the 1+1 Haar measure is flat in rapidity). Sprinkled causal sets gain about $\ln 2$ links per element when N doubles, because links extend along the light cone [3]. The rule reproduces this. For $N = 1\text{k}, 2\text{k}, 4\text{k}$ and $c = 1$ the mean link count per element is $5.17 \to 5.95 \to 6.59$, against $5.49 \to 6.18 \to 6.87$ for sprinkling, and the interval abundances $N_1/N_0, N_2/N_0, N_3/N_0$ agree with sprinkling to ± 0.02 for $c = 1$–3. A rule that is local in a spatial slice instead gives $2.92 \to 3.05 \to 3.08$ links and abundances 0.96–0.97: it is not manifoldlike.

A placement field test confirms frame independence. We apply Sorkin's smeared d'Alembertian B_ϵ [12] ($\epsilon = 0.1$) to plane waves of fixed invariant wavenumber and varying rapidity b. The response ratio $R(b)/R(0)$ for the densifying network stays within 1.02–1.06 for $|b| \leq 2.5$, as for Poisson sprinkling, at about 5% precision. A variant that replicates only the leaves of the genealogy ("replication of the present") retains a 15–30% frame dependence that does not shrink from $N = 6\text{k}$ to 12k. The reason is that the Lorentz group is non-compact: a single splitting can only draw rapidities from a bounded range, so the genealogy keeps a preferred frame. Inserting into the existing interior lets the usable rapidity range grow with N.

B The 3+1-dimensional rule

In four dimensions the child is placed at proper time and rapidity

$$\tau = \ell \left(\frac{24E}{\pi}\right)^{1/4}, \quad E \sim \mathrm{Exp}(1), \qquad p(\eta)\, d\eta \propto \sinh^2\eta\, e^{-\alpha(\cosh\eta - 1)}\, d\eta,$$

with $\Delta t = \pm\tau\cosh\eta$, $|\Delta\mathbf{x}| = \tau\sinh\eta$ and a uniformly random spatial direction. Here $(\pi/24)\tau^4$ is the volume of the parent–child diamond, so E is its expected element count in units of the discreteness density. The sign of Δt is chosen with probability 1/2 (time-symmetric insertion) and children falling outside the region are redrawn.

Two features have no 1+1 analogue. First, insertion must be time-symmetric. In 1+1 dimensions every insertion in rank coordinates is a monotone reparametrisation of u and v, i.e. a conformal map, and the infinite-dimensional conformal group absorbs it. In 3+1 dimensions it does not; future-only growth collapses toward the final boundary (1584 of 2000 elements in the last sixth of the time range). Second, the Haar measure is not normalisable. Without damping ($\alpha = 0$) the median spatial step grows as $N^{1/4}$, i.e. in proportion to the region size, and the large-scale density is set by the boundary (the diamond tips are over-dense by a factor 2.4–3.2, independent of N from 2k to 32k).

The local structure is nevertheless Lorentz invariant. Our order-only *diamond test* selects parent–child pairs whose interval contains $60 \pm 10\%$ elements, measures the Myrheim–Meyer dimension, links per element, longest chain and N_1/N_0 inside, and bins by the rapidity b of the separation. For $N = 32\mathrm{k}$ the change between $b \in [1.2, 1.6)$ and $b \in [0, 0.4)$ is within one standard error for Haar replication (e.g. $\Delta d_{\mathrm{MM}} = +0.014$, 0.9 s.e.) and for Poisson sprinkling, while a lattice control drifts by 12–15 s.e.

C Damping, the state-determined frame, and what replication cannot do

Damping by $e^{-\alpha\cosh\eta}$ cures the boundary pile-up at the cost of a frame u in which $\cosh\eta$ is measured. We define u from the state: $u(p)$ is the normalised sum of future unit vectors to the K related elements of smallest interval volume, which minimises $\sum_i \cosh\eta_i$ and is therefore the rest frame of the neighbour "gas". Boosting a whole set by $b = 0.5$ and 1.0 returns estimates 0.51 and 1.01 (spread 0.1 for $K = 100$), and during growth u does not self-amplify. In a periodic time slab ($T = 0.5$, spatial torus of side 1) the edge-to-centre density ratio is 3.75 (8k) and 2.62 (32k) without damping and 1.82 and 1.08 with $\alpha = 1$. (This slab ratio drifts with N because the Perron vector of the time kernel drifts as the step shrinks, Sec. IV A; it is a different quantity from the N-independent tip over-density of the diamond in Sec. II B.) For $\alpha \ll 1$ the mean rapidity factor is $\langle\cosh\eta\rangle = K_2(\alpha)/K_1(\alpha) \approx 2/\alpha$, so the step length scales as $\ell/\alpha = L_H$.

Replication is a kinematics, not a dynamics, and two negative results fix what it cannot do. With only temporal boundaries (a periodic spatial torus), replication produces a time-dependent conformal factor, i.e. a spatially flat FRW geometry $a(t)^2\eta$ with $a = (\rho/\bar{\rho})^{1/4}$. But its shape is a U-shaped bounce whose total variation is a factor 1.1–1.5, fixed by the aspect ratio of the box. Pure future replication has no steady state: a mean-field analysis shows that near the future boundary the generation rate always exceeds the local density, for every normalisable profile. Weighting configurations by a Euclideanised action $e^{-\beta S}$ also fails: at $N = 1500$ we find a uniform continuum-like phase for $-0.05 \le \beta \le 0.1$, the known layered phase for $\beta \le -0.1$ and the bilayer phase for $\beta \ge 0.3$, never an expanding profile. Meanwhile the smeared BD action of replicated sets equals that of matched Poisson controls to within -0.013 ± 0.010 per element against a volume term of 0.316. Expansion must therefore come from the Lorentzian path sum; since the BD action on these sets reproduces the Einstein–Hilbert action within resolution, we expect its semiclassical saddle

to be the Friedmann solution, but we have not shown this; replication supplies the microscopic, Lorentz-invariant kinematics.

III Lorentz invariance versus locality

The model meets the Lorentz–locality dilemma in three distinct forms. Each is a trade-off between a local property and a property that requires mixing over the light cone, and in each case the model keeps Lorentz invariance at the discreteness scale and pays elsewhere.

Spatial locality versus frame independence. A child placed near its parent in a spatial slice (Sec. II A) produces a non-manifoldlike order. A child placed at small proper time but uniformly random rapidity produces an order indistinguishable from sprinkling, and the parent–child pair is a causal nearest neighbour: 34% of pairs are directly linked for $c = 1$. Locality survives in its Lorentz-invariant form, closeness in proper time, and is lost in its frame-dependent form.

Frame independence versus spatial coherence. A field inherited along the genealogy is spatially incoherent on the densifying network: its equal-time correlation is almost flat, because the genealogy has depth only $\ln N$ and children travel far along the light cone. Writing replication as an entangling operation shows that the conflict is exact. If each element is a bosonic mode and replication applies a SUM gate to the parent and a fresh vacuum mode, the resulting Gaussian state has equal power per octave, $\langle |q_k|^2 \rangle \propto k^{-z}$ with $z = d$; we measure $z = 1.03$ on 2048 modes in one dimension and $z = 1.93$ on 4096 modes in two. In three dimensions this is the $z = 3$ anisotropic scaling, not the Lorentz-invariant $z = 1$. Coherence must therefore come from field dynamics on the Lorentz-invariant geometry, with a horizon separating the vacuum at short distances from inherited correlations at long distances.

Uniformity versus locality. Damped replication is a measure-valued Pólya urn: a region's production rate is proportional to its current density, so fluctuations are frozen in rather than averaged out. At $N = 128$k with $\alpha = 1$, the Fano factor of diamond counts rises from 1.48 at $E \approx 30$ to 4.58 at $E \approx 620$, while the matched Poisson control stays at 0.92–1.07. The longest-chain statistic $L/n^{1/4}$ is unaffected (2.107 against 2.133 for sprinkling). Uniformity needs nonlocal mixing, and nonlocal mixing lets the boundary dominate (Sec. II B). Section IV shows that clumping is confined to scales above the damping length, which is the Hubble scale physically.

A Where Lorentz violation enters

Mean counts are exactly boost independent: Sec. IV shows that the expected count in any diamond is its conformal volume. Lorentz violation enters only through the clumping excess. Over 3000 diamonds with rapidities up to 1.65, the excess Fano factor $F - 1$ falls by about 20% per unit rapidity ($d \ln(F-1)/d\eta = -0.19$ for $\alpha = 1$, -0.22 for $\alpha = 3$). For chains the effect is an order of magnitude smaller, and for $\alpha \leq 0.3$ the frame dependence of the chain constant is below 0.1% (Sec. IV C). In the physical regime the sub-Hubble clumping excess is of order $(H/M_{\mathrm{Pl}})^2$, so the violation is unobservable.

Uniform bounds over all diamonds of a bounded region pay only a logarithmic price for the non-compactness of the Lorentz group. In a region of duration T, diamonds of volume at least $V_{\min}$ automatically satisfy $u \geq \tau_{\min}^2/2T$ for their short null side u, so the maximal rapidity is about $\log(T/\tau)$. This is the same geometric fact that Boguñá and Krioukov identify as the non-amenability of rapidity space [16]; our state-determined Maxwell–Jüttner damping realises a dynamical version of their fixed rapidity cut-off. Defining the cut-off by $\alpha \cosh \chi_c = 1$ (damping by $1/e$) gives $\chi_c = \operatorname{arccosh}(L_H/\ell) \approx \ln(2L_H/\ell)$.

IV Mesoscopic manifoldlikeness: F2 and F3

Madsen has shown that a causal set admits an essentially unique embedding into a spacetime if it satisfies three conditions on mesoscopic causal diamonds [17]: order preservation (F1), counts matching volume within a tolerance that shrinks as the volume grows (F2), and longest chains matching proper time (F3). F1 holds by construction for the replication model. This section proves high-probability versions of F2 and F3 for the conformal metric $g = a^2\eta$, where $a = (\rho_n/\bar{\rho})^{1/4}$ is set by the time profile of the density. Clumping is the only obstruction, and it is confined to scales above the damping length.

A Exact second moments and the clumping criterion

Consider first the spatial process on the torus $\mathbb{T}^d$: n_0 independent uniform seeds, then each step adds a child $X_{n+1} = X_J + \xi$ with J uniform among the n existing points and ξ i.i.d. symmetric with characteristic function $\varphi(k)$. Let $S_n(k) = \sum_j e^{ik\cdot X_j}$.

Proposition 1 (exact second moment). $A_n(k) = \mathbb{E}|S_n(k)|^2$ satisfies $A_{n+1} = A_n\,(1 + 2\varphi(k)/n) + 1$ with $A_{n_0} = n_0$, hence

$$A_n(k) = \frac{\Gamma(n+2\varphi)}{\Gamma(n)}\left[\frac{n_0\,\Gamma(n_0)}{\Gamma(n_0+2\varphi)} + \sum_{m=n_0}^{n-1}\frac{\Gamma(m+1)}{\Gamma(m+1+2\varphi)}\right], \qquad \varphi = \varphi(k).$$

Proof. $|S_{n+1}|^2 = |S_n|^2 + 1 + 2\,\mathrm{Re}(\bar{S}_n e^{ik\cdot X_{n+1}})$ and $\mathbb{E}[e^{ik\cdot X_{n+1}} \mid \mathcal{F}_n] = \varphi(k)S_n/n$. □

With $R_n = A_n/n$ (Poisson = 1), modes with $\varphi < 1/2$ saturate at $R_n \to 1/(1-2\varphi)$, while modes with $\varphi > 1/2$ grow without bound relative to shot noise, $R_n \simeq c(\varphi)\,n_0^{1-2\varphi}n^{2\varphi-1}$. The threshold $\varphi(k) = 1/2$ is the clumping criterion; for a Gaussian step of width s it sits at $|k| = \sqrt{2\ln 2}/s \approx 1.18/s$. The window Fano factor follows exactly: $F_W = |W|^{-1}\sum_{k\neq 0}|c_k|^2 R_n(k) \le 1 + \delta_{\mathrm{sub}} + E_W\sigma^2_{\mathrm{LS}}$, where c_k are Fourier coefficients of the window, δ_{sub} comes from sub-step modes and σ^2_{LS} from the amplified large-scale modes. In simulations (3D, $s = 0.05$, $n = 2\times 10^4$) the predicted R_n and window Fano factors agree with measurement to 1–2.5 standard errors.

Proposition 2 (time-resolved second moment). For the 4D process with a time boundary, let $F_n[i]$ be the Fourier mode restricted to time bin i, $C_n = \mathbb{E}[F_nF_n^H]$, ρ_n the expected bin counts, $Q_n(k)[i,i'] = \mathbb{E}[\mathbf{1}(\text{child in } i)\,e^{ik\cdot\Delta x} \mid \text{parent in } i']$ and $P_n = Q_n(0)$. Then

$$\rho_{n+1} = \rho_n + \tfrac{1}{n}P_n\rho_n, \qquad C_{n+1} = C_n + \tfrac{1}{n}\left(Q_nC_n + C_nQ_n^H\right) + \tfrac{1}{n}\,\mathrm{diag}(P_n\rho_n).$$

The $k = 0$ part fixes the time profile (the Perron vector of P_n) and the $k \neq 0$ part fixes the clumping spectrum. A mode grows relative to shot noise iff $\lambda_{\max}(Q_n(k)) > 1/2$. Without fitted parameters, the recursion reproduces 17 of 18 simulated $R(k)$ values within 2σ (12 seeds, $\alpha = 0, 1, 3$) and the time profiles to ± 0.07.

Proposition 3 (eigenvalue squeeze). With r the spatial step, $\lambda_{\max}(Q(k)) \le \max_{\text{parent}}\mathbb{E}|\mathrm{sinc}(kr)|$ always, and $\lambda_{\max}(Q(k)) \ge \Phi(k) = \mathbb{E}\,\mathrm{sinc}(kr)$ in the time-translation-invariant bulk. For damped Haar steps with $\alpha \ll 1$, $\cosh\eta$ is Gamma$(2, 1/\alpha)$-distributed and, with $\kappa = k\tau/\alpha$,

$$\Phi = \frac{1}{1+\kappa^2}, \qquad \mathbb{E}|\mathrm{sinc}| = \frac{\coth(\pi/2\kappa)}{1+\kappa^2}.$$

Modes with $\kappa \le 1$ must clump and modes with $\kappa \ge 1.122$ cannot. Substituting $\tau = \ell(24E/\pi)^{1/4}$ and $\alpha = \ell/L_H$ eliminates ℓ, and averaging over E gives a critical wavenumber $k_cL_H \in [0.689, 0.785]$: clumping starts at wavelengths between 8.0 and 9.1 damping lengths.

For space-time modes $e^{i(k\cdot x - \omega t)}$, with $\nu = \omega\tau/\alpha$, the bulk amplification factor is

$$\Phi(\kappa,\nu)=\frac{1}{2\kappa}\left[\frac{\kappa+\nu}{1+(\kappa+\nu)^2}+\frac{\kappa-\nu}{1+(\kappa-\nu)^2}\right],$$

which reduces to $1/(1+\kappa^2)$ at $\nu=0$ and to $(1-\nu^2)/(1+\nu^2)^2$ at $\kappa=0$. At fixed τ, pure temporal clumping starts at $\nu_c=\sqrt{\sqrt{5}-2}=0.486$. Averaging over E exactly as for the spatial modes, with $\nu=\omega L_H(24E/\pi)^{1/4}$, moves the onset to $\omega_c L_H=0.326$: temporal clumping occurs only for periods above $19.3\,L_H$ (Proposition 7 treats this temporal sector, where non-Perron modes of P with eigenvalue above $1/2$ form a Pólya urn).

B Concentration and uniform F2

For a test function f let $N_f=\sum_{j\le n} f(X_j)$ and $G_l=(I+K_l/l)\cdots(I+K_{n-1}/(n-1))$, where K_m is the child kernel at population m. Then $M_l=\sum_{j\le l}(G_l f)(X_j)$ is a martingale with $M_n=N_f$, because $\mathbb{E}[g(X_{l+1})\mid\mathcal{F}_l]=\mu_l(K_l g)$.

Theorem 4D (oscillation–Azuma). For bounded f,

$$\mathbb{P}\left(|N_f-\mathbb{E}N_f|\ge t\right)\le 2\exp\!\left(-\frac{2t^2}{V_f}\right),\qquad V_f=n_0\,\mathrm{osc}(G_{n_0}f)^2+\sum_{l=n_0}^{n-1}\mathrm{osc}(G_{l+1}f)^2.$$

Using the oscillation instead of $2\|\cdot\|_\infty$ removes the Perron (constant) direction automatically. **Theorem 5D (Freedman form)** sharpens this to $2\exp(-t^2/2(\bar\sigma^2+bt/3))$ on the event $\{\langle M\rangle\le\bar\sigma^2\}$ with $b=\max_l \mathrm{osc}(G_{l+1}f)$ [18], and **Lemma 6D** shows by bootstrapping Theorem 4D over Fourier–cosine modes that this event has probability at least $1-\delta$ with $\bar\sigma^2=\sum_l w_l(1+\varepsilon_l)$, where w_l is the variance of $G_{l+1}f$ under the child marginal and ε_l is $\sqrt{\log(1/\delta)}$ times the large-scale density contrast at the window scale. Numerically the bound $\sum_l \mu_l(u_l)$ on $\langle M\rangle$ exceeds the exact quadratic variation by less than 0.2%.

Theorem 8 (uniform F2 over diamonds). Let $\mathcal{D}(u_{\min})$ be all causal diamonds in the region with short null side at least $u_{\min}$. Diamonds are bracketed by grid-point pairs, $I(p^+,q^-)\subset I(p,q)\subset I(p^-,q^+)$ exactly, and Theorem 5D is applied to every bracket. With probability at least $1-\delta-\delta'$, simultaneously for all $D\in\mathcal{D}(u_{\min})$,

$$\left|\frac{N_D}{E_D}-1\right|\le\sqrt{\frac{2(1+\varepsilon)F_D L}{E_D}}+\frac{2b_D L}{3E_D}+\sqrt{\frac{F_D}{E_D}},\qquad E_D=\bar\rho\,\mathrm{Vol}_g(D),$$

with $L=O(\log N+\log(T/u_{\min})+\log 1/\delta)$. Below the clumping scale F_D and b_D are $O(1)$ and the tolerance is Poissonian up to logarithms, which is the form F2 requires. In simulations ($N=128$k, 100 realisations, 3000 diamonds with $E=30$–800 and rapidities up to 1.65) the mean count divided by the conformal volume is 0.9997 ($\alpha=1$) and 1.0014 ($\alpha=3$), while the flat volume is off by -15% to $+35\%$. The supremum of the standardised deviation over M' random diamonds grows like $\sqrt{2\ln 2M'}$, the signature of a logarithmic uniformisation cost.

Theorem 16 (the price of uniformity is universal). For a Poisson sprinkling, 500 pairwise disjoint diamonds with $E=30$ already force $\mathbb{P}(\sup Z\ge 2.73)\ge 0.95$ (with probability 0.957 some diamond holds at least 45 points, $Z=2.739$; the median of $\sup Z$ is 3.29), and a Sudakov argument in the Gaussian limit gives $\mathbb{E}\sup Z\ge 4.15$. The measured continuous supremum is 7.6 ± 0.4 standard deviations for Poisson sprinkling and 8.1 ± 0.5 for the replication process; Theorem 8 gives 20–23. Clumping therefore adds only about 0.5σ to a cost every sprinkling pays, and Theorem 8 is loose by a factor of about 2.7.

C F3: longest chains

Write the process as a function of independent inputs (seeds, then parent index and displacement seed for each step), revealed in growth order. Resampling input m moves only the point it created and that point's descendants S_m, so a maximal chain C loses at most $|C \cap S_m|$ elements.

Theorem 9 (Doob-martingale variance bound). $\operatorname{Var} L_D \le 2\,\mathbb{E}\,\Gamma_D^{\mathrm{piv}} \le 2\,\mathbb{E}\,\Gamma_D$, where $\Gamma_D^{\mathrm{piv}} = \sum_m \min_C |C \cap S_m|^2$ and $\Gamma_D = \sum_m |C \cap S_m|^2$ for a fixed maximal chain C, which counts shared ancestry along the chain. Its self part is L times the genealogical depth, $1 + \ln(n/n_0)$ for a uniform recursive tree. **Theorem 10** upgrades this to exponential tails by Talagrand's convex-distance inequality [19]: conditionally on the first m^* inputs, $\mathbb{P}(L \ge s + t,\ \Gamma_{>m^*} \le \gamma)\,\mathbb{P}(L \le s) \le e^{-t^2/4\gamma}$. **Lemmas 11 and 12** bound the certificate size by the maximal family size times the depth times L, with a priori tails for both, so the tail bound uses no measured input.

Theorem 13 and Proposition 15 (local Poissonisation). Conditionally on a prefix, the new points form a superposition of independent families, a Poisson cluster process in the Athreya–Karlin continuous-time embedding [20]. The process is not close to Poisson in total variation, since a diamond typically contains about 50 families with two or more points. But an exact inclusion–exclusion expansion shows that the mean longest chain differs from its Poisson value only through intra-family correlations weighted by chain interaction kernels, which are tiny. The leading pair kernel satisfies $\Psi_2 \le 0$ for spacelike pairs and, by a Russo identity, averages to $\ell''(E) \approx -(3/16)\ell/E^2$. The surviving deviation comes from family chains of three or more points and renormalises the chain constant by $\delta_{\mathrm{cl}} \approx 0.1$–$0.3\%$ at $\alpha = 3$. For $\alpha \le 0.3$ both this renormalisation and its frame dependence vanish below 0.1%.

Numerically ($N = 128$k, 800 diamonds, 100 realisations) the mean longest chain equals that of a Poisson sprinkling with the same time profile to 0.04–0.12% for $\alpha = 1$, and its variance exceeds the Poisson value by only 3–5%, although the count Fano factors of the same diamonds are 1.3–7.7. Chains are thin one-dimensional objects with many near-maximal competitors, so F3 is far more robust than F2. Hence $L \approx m_4(\bar{\rho}\,\mathrm{Vol}_g)^{1/4}$, with m_4 the four-dimensional chain constant of Bollobás and Brightwell, known only to satisfy $1.8554 < m_4 < 2.5297$ [21–23].

V Continuum limit: poset limits under interior insertion

Interior insertion changes what a continuum limit can mean. Bento, Dowker and Zalel accept that the order of birth is gauge and allow elements to be born in the past, but keep the condition that no element is born between two earlier ones, so that finite convex suborders eventually settle and serve as observables [15]. The replication process violates exactly this condition: each stage is an induced suborder of the next (old relations never change), but no finite convex suborder ever settles. We therefore use Janson's induced-subposet densities $t(Q, P_N)$, the probability that $|Q|$ uniformly chosen elements of P_N induce the order Q [24]. Growth is read as refinement and N as resolution. The cost is the loss of a settled past, one step beyond the position of [15].

A A conditional limit theorem

Theorem 17 (poset limit). Let μ_N be the empirical measure of the first N elements in the region $\bar{\mathcal{M}}$. Suppose (H1) $\mu_N \Rightarrow \mu_\infty$ almost surely and (H2) $\mu_\infty \otimes \mu_\infty$ assigns zero mass to null-separated and coincident pairs. Then for every finite poset Q, almost surely

$$t(Q, P_N) \longrightarrow t(Q, W_{\mu_\infty}), \qquad W(x, y) = \mathbf{1}[x \prec y],$$

the density of Q in the strict-order kernel of μ_∞.

Proof. $t(Q, P_N) = \int \mathbf{1}_Q \, d\mu_N^{\otimes k} + O(k^2/N)$. Product measures of weakly convergent sequences converge weakly. The indicator $\mathbf{1}_Q$ is discontinuous only where some pair of arguments is null-separated or coincident, a $\mu_\infty^{\otimes k}$-null set by (H2), so the portmanteau theorem applies. □

Discreteness-scale clumping affects only point pairs at separations of order ℓ, whose measure vanishes in the limit; Theorem 17 is blind to it.

B The undamped case: an explicit limit

Theorem 18 ((H1) for $\alpha = 0$). For $\alpha = 0$ in the slab $(0, T) \times \mathbb{T}^3$, $\mu_N \Rightarrow \nu$ almost surely, where

$$\nu = \nu_T(t)\, dt \otimes \mathrm{Unif}(\mathbb{T}^3), \qquad \nu_T(t) = \frac{3\left(t^2 + (T-t)^2\right)}{2T^3}.$$

Since ν has a bounded density, (H2) holds automatically and Theorem 17 applies unconditionally: the poset limit is a uniform sprinkling of $(\mathcal{M}, a(t)^2\eta)$ with $a^4 \propto t^2 + (T-t)^2$, and it is deterministic.

As the step scale $\tau \to 0$ the undamped kernel becomes macroscopic and null. Its time part is $q(t'|t) = |t' - t|/Z(t)$ with $Z(t) = (t^2 + (T-t)^2)/2$, and its spatial part is a displacement of length $|\Delta t|$ in a uniform direction. Since $q(t'|t)Z(t)$ is symmetric, the time chain is reversible with respect to $Z \propto \nu_T$, so $\nu K = \nu$. Three lemmas drive the proof.

- **L1 (Doeblin).** $K^5(x, \cdot) \geq c\,\lambda$ with $c \geq 1.37 \times 10^{-6}$ and λ uniform on a central slab, uniformly in x.
- **L2 (smoothing).** $\mathrm{TV}(K^4(x, \cdot), K^4(x', \cdot)) \leq 436\,|x - x'|$.
- **L3 (kernel perturbation).** $|(K_m - K)h| \leq C\bar{\tau}_m(\|h\|_\infty + \mathrm{Lip}\, h)$ with $\bar{\tau}_m \propto m^{-1/4}$.

L1 and L2 rest on two explicit facts. The convolution of uniform spherical shells of radii a, b, c in $\mathbb{R}^3$ has the rotation-invariant density $h_{abc}(|x|)$ on $\mathbb{R}^3$ with $h_{abc}(s) = \ell(s)/(16\pi abc\, s)$, where $\ell(s)$ is the length of $[|s-c|, s+c] \cap [|a-b|, a+b]$ (the density of the radius s itself is $4\pi s^2 h_{abc} = s\,\ell(s)/4abc$); it lies in $W^{1,1}$ with $\|\nabla h_{abc}\|_1 \leq (a+b+c)^2/2abc$. And because the time step ρ has density proportional to ρ (the Haar weight), small radii are rare: $\mathbb{E}[1/\rho \mid t] = T/Z(t) \leq 4/T$. All constants here and in L1–L2 are for $T = 1/2$ and a spatial torus of side 1, so $T/Z \leq 8$ and $\int_0^T Z\, dt = T^3/3 = 1/24$. The proof then writes $\mu_{m+1} = \mu_m A_m + \Delta_{m+1}/(m+1)$ with $A_m = I + (K_m - I)/(m+1)$ and Δ a martingale-difference measure, controls the homogeneous part by L1, the perturbation by L2–L3 (early perturbations are forgotten by mixing), and the noise by a martingale bound along geometric subsequences with Borel–Cantelli. The ideal kernel predicts an order fraction $r = 0.0679$; simulations of the ideal kernel give 0.0678–0.0682 from $N = 8$k on, and the time profiles agree to 0.01. The finite-τ process approaches this limit as $\tau \propto N^{-1/4}$ shrinks. The setting is an infinitely-many-colour Pólya urn [25, 26]; what is new here is the time-inhomogeneous kernel K_m, which is mutually singular with its limit K.

C The damped case: a random limit

Theorem 19 ((H1) for $\alpha > 0$). $\mu_N \Rightarrow \mu_\infty$ almost surely, and for Lipschitz f the drift part of $|\mu_N(f) - \mu_\infty(f)|$ is at most $\mathrm{Lip}(f)\, C_\alpha N^{-1/4}$ with $C_\alpha \propto K_2(\alpha)/K_1(\alpha) \approx 2/\alpha$.

The mean step is bounded by $\sqrt{2}\,\Gamma(5/4)\,(24V/\pi m)^{1/4} K_2(\alpha)/K_1(\alpha)$, so the drift is summable, $\sum_m m^{-5/4} < \infty$, while the martingale part is bounded in L^2. The asymptotic regime requires $N \gg \alpha^{-4}$, which explains the slow numerical convergence for small α. Unlike the case $\alpha = 0$, the limit is random: it is set by the early history, as in a Pólya urn.

Theorem 20 ((H2) for $\alpha > 0$). $\mu_\infty \otimes \mu_\infty(\mathcal{Z}) = 0$ almost surely, where $\mathcal{Z}$ is the set of null-separated or coincident pairs.

Proof sketch. By the portmanteau theorem for the open set $\mathcal{Z}_\varepsilon = \{||\Delta t| - |\Delta y|| < \varepsilon\}$ and Fatou's lemma it suffices to show $\sup_N \mathbb{E}[\mu_N \otimes \mu_N(\mathcal{Z}_\varepsilon)] \to 0$. Acceptance and redraws depend only on time, so the time process is autonomous and, given it and the genealogy, positions perform a translation-equivariant random walk on the tree. The single-step displacement density is bounded because the factor τ^3 from $E \sim \mathrm{Exp}(1)$ cancels the hyperbolic Jacobian $\tau^3 \sinh^2 \eta$. For a pair of points, take the earliest step b^* below their most recent common ancestor; the pair lands in $\mathcal{Z}_\varepsilon$ with probability at most $C_\alpha \varepsilon b^*$. Genealogical compactness gives the exact Pólya bound $\mathbb{P}(b^* > M) = 2(N-M)/((M+1)(N-1)) \le 2/(M+1)$. Choosing $M = \lceil \varepsilon^{-1/2} \rceil$ yields $\mathbb{E}[\mu_N \otimes \mu_N(\mathcal{Z}_\varepsilon)] \le (2 + 2C_\alpha)\sqrt{\varepsilon} + 1/N$. □

Corollary. For every $\alpha \ge 0$ the replication process has a poset limit, the uniform sprinkling of the conformally flat, spatially homogeneous metric $a(t)^2\eta$ with $a^4 \propto \rho_\infty(t)$.

D Numerical evidence

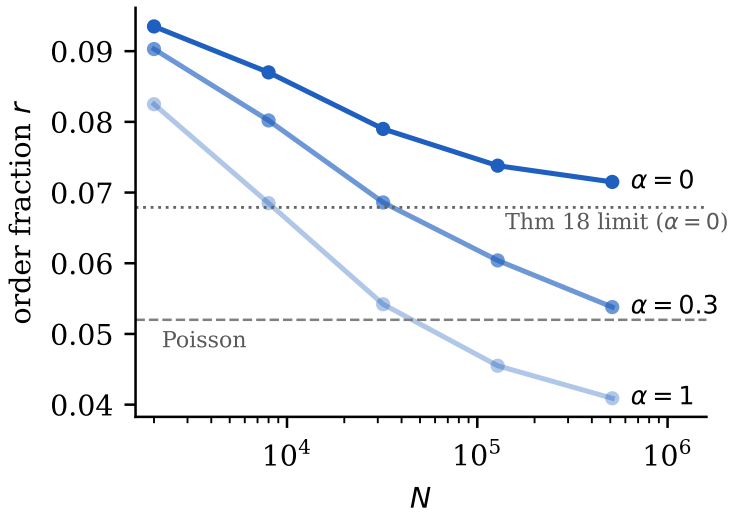


Figure 1: Order fraction $r = t(2\text{-chain})$ of the 4D replication process along one growth path, against N, for $\alpha = 0$, 0.3 and 1; the dotted line is uniform Poisson sprinkling. Monte Carlo error about 4×10^{-4}.

The order fraction approaches its limit slowly, as Theorem 19 predicts. For $\alpha = 0$ the decrement per doubling of N falls from 0.0048 to 0.0007; a naive extrapolation of these decrements gives about 0.070, 3% above the Theorem 18 value 0.0679, the difference being the slowly vanishing finite-τ correction ($\tau \propto N^{-1/4}$); for $\alpha = 1$ the extrapolated limit is about 0.038, while $\alpha = 0.3$ is still pre-asymptotic at $N = 512$k. The limit is set entirely by the time profile: keeping the times and redrawing the spatial positions uniformly reproduces the order fraction (0.0739 against 0.0739, 0.0601 against 0.0602, 0.0454 against 0.0456), whereas keeping the positions and redrawing times uniformly returns the Poisson value. Smoothing the point cloud below the discreteness length (kernel width 0.01 against $\ell = 0.044$) changes every three-element density by at most 6×10^{-4}, equal to the Monte Carlo error, which confirms that the limit does not see discreteness-scale clumping. Across four realisations the densities vary by at most 6×10^{-4}, far less than their distance to Poisson values.

The abstract measure-theoretic layer of Theorems 17–20 is machine-checked in Lean (Sec. VIII).

VI Rogue posets and the failure of order-only rules

A dynamics that works in coordinates must still be judged by its order alone. We therefore built an intrinsic test battery that reads only the relation matrix, and used it to ask whether the replication rule can be replaced by a rule that never sees coordinates.

A The intrinsic battery

Intervals are selected by element count, so every test is invariant. The battery measures the Myrheim–Meyer dimension from the order fraction [27, 28], the midpoint dimension $\log_2(n/n_{\rm half})$, the link fraction, the interval-abundance spectrum N_k/R (pairs with exactly k elements between them) [29], the 2D BD action per element, the height and the fraction of minimal elements. We add one statistic that proved decisive: the spread of the Myrheim–Meyer dimension across intervals. Calibration shows why several tests are needed. Transitive percolation at $p = 0.05$ fools the dimension estimator ($d_{\rm MM} = 1.93$) but is exposed by the abundance spectrum, which peaks at $k = 1$–2, and by the BD action ($S/n = +3.43$). KR orders fail every test. Output of the replication model passes all of them: in 2D, deviations from flat sprinkling range from 1% ($\alpha = 0$) to 5% ($\alpha = 3$), and re-sprinkling with the grown density field reproduces 60–80% of that deviation, so most of it is the conformal factor rather than microstructure.

In 2D the rule can be made fully intrinsic. A 2D order is determined by its realizer, a pair of linear orders, up to symmetry, so the realizer can serve as ordinal light-cone coordinates. Placing the child with $\Delta u\,\Delta v = m\,x$, $x \sim \mathrm{Exp}(1)$, and $\Delta u/\Delta v = e^{2\eta}$ (one element's proper volume, uniform rapidity) passes the battery. A naive kernel that keeps both rank displacements small instead grows filamentary chains ($d_{\rm MM} = 1.15$–1.7), a direct order-theoretic image of Sec. III. The realizer trick does not extend to 4D, because Minkowski sprinklings have no finite realizer dimension. This is the core of the rogue-poset problem in four dimensions.

B Three families of order-only insertion, three failures

An order-only insertion chooses a down-set D and an up-set U with every element of D below every element of U, and makes the new element the cut between them. Its past and future cannot be inferred from each other. We tested every natural way of choosing them; each family fails in a characteristic way.

Family	Representative rule	2D outcome ($N = 2000$; sprinkling: $d_{\rm MM} = 2.00$, link fraction 0.07)	Failure
Closure	Galois closure: future = all upper bounds of the chosen past	Height 510–617 against 79; $d_{\rm MM} = 1.30$–1.37; $S/n \approx 1.00$	Too many relations; positive feedback grows chains
Minimal cuts	Past $= J^-[x]$, future $= J^+[b]$ for a small interval $I(x, b)$	$d_{\rm MM} = 2.9$–3.2	Too few relations
Link-level mutation	Clone a parent, drop links with probability μ, add siblings with probability γ	Mean quantities tunable to a few per cent; spread of $d_{\rm MM}$ across intervals 0.20–0.28 against 0.05; 2–3% exact twins	Fluctuations; parameters not portable between dimensions

Family	Representative rule	2D outcome ($N = 2000$; sprinkling: $d_{\mathrm{MM}} = 2.00$, link fraction 0.07)	Failure
Element-level moves	Clone, then single-element boundary moves to a volume-controlled displacement	$d_{\mathrm{MM}} = 2.5$–3.9; link fraction 0.26–0.62	Light cones become jagged

The closure failure persists even with an oracle. If the new element receives its true geometric past and only its future comes from the closure, spurious relations per insertion grow from a median of 3 early to 17 late, and the height grows from 79 to 191. The bias is one-sided, because the true future is always contained in the closure, so it accumulates. In static 4D sprinklings the closure adds about 0.3 spurious relations per past link, growing as roughly $\sqrt{N}$ (26, 50 and 99 at $N = 2$k, 8k, 32k), whereas in 2D it stays near 2.

The jagged-cone failure is structural, not a matter of counts. Forcing the relation count to be conserved leaves the outcome unchanged; every boundary move links the new element to some arbitrary boundary element, so the cone loses coherence. Larger displacements make it worse: at three times the nearest-neighbour displacement the link fraction is eight times the sprinkled value.

C The mechanism: order-space neighbours are far apart

All three failures have one cause. Define the order distance $\Delta(c, y)$ as the number of elements related differently to c and to y. In a sprinkling, the median distance to the order-nearest neighbour is 18, 26, 38 at $N = 1$k, 2k, 4k in 2D, about $0.59\sqrt{N}$, and 20, 38, 68 in 4D, about $N^{0.8}$. The thin shell between the light cones of two nearby points extends along null directions to the boundary of the region, so no displacement of $O(1)$ elements exists. A rule that edits relations locally cannot move a light cone coherently. A geometric displacement does exactly that, which is why the successful rules (the 2D realizer rule and the coordinate model) carry structure beyond the order. The same geometric fact underlies the growth of closure errors and the non-amenability noted in Sec. III.

We conclude that the coordinate model is the dynamics, while the intrinsic battery and the poset-limit theorems are verification tools. The answer to the rogue-poset problem offered here is that the coordinate model's output passes every intrinsic test, and that order-only local rules necessarily produce jagged or rogue structure. Section VII asks whether quantum interference can remove that structure after the fact.

VII Quantum dynamics: interference and the defect gas

Section VI leaves a precise question. Order-only growth rules produce non-manifold structure of specific kinds: near-chains from closure, jagged cones from element moves, clone families from mutation. Attaching the phase $e^{i\beta S_{\mathrm{BD}}}$ to the histories of such a rule, does interference remove the non-manifold branches?

A Phases on a growth tree telescope

We follow the path-sum convention in which the amplitude of an event is the coherent sum over its members [9, 10, 30], within Sorkin's quantum-measure framework [8]. If every growth step carries the phase $e^{i\beta\Delta S}$, the phase of a history is $e^{i\beta(S(C_N)-S(C_1))}$, so all histories ending in the same poset are in phase. The growth tree only supplies weights $w(C) = \sum_h \sqrt{p(h)}$, and interference happens

only between different final posets. A decoherence functional that is diagonal in final posets would make the phase irrelevant altogether.

A class Γ of posets is not suppressed merely because its measure falls below its $\beta = 0$ value: random phases already reduce every class to the "random floor" $\sqrt{\sum w^2}$, which favours small classes relative to large ones. The right diagnostic is the excess cancellation

$$E_\Gamma(\beta) = \frac{\left|\sum_{C\in\Gamma} w(C)\, e^{i\beta S(C)}\right|^2}{\sum_{C\in\Gamma} w(C)^2},$$

equal to 1 for random phases, below 1 for systematic cancellation and above 1 for partial coherence.

B Exact enumeration and variance scaling

Enumerating all posets with $N \leq 8$ (matching OEIS A000112 and A001035), the geometric mean of E_Γ over $\beta \in (0, \pi]$ for the 2D action is 0.09, 0.03, 0.01 at $N = 6, 7, 8$ for two-layer posets: they cancel systematically, increasingly with N, reproducing at small N the Loomis–Carlip mechanism [9]. Three-layer (KR-type) posets sit at the random level (0.88–1.30), as expected at sizes far from the asymptotic regime of [10]. 2D orders are partially coherent (1.42, 3.20, 6.27), because their actions are relatively concentrated.

For $N = 50$–400 we sample each class 300 times and measure the scaling of $\mathrm{Var}(S)$. With the unsmeared 2D action, sprinkled causal sets have $\mathrm{Var} \propto N^{1.42}$, clone families $N^{1.59}$, KR orders $N^{2.04}$ and jagged-cone orders $N^{3.11}$; near-chains from closure have a variance a thousand times smaller than sprinkling, because an n-chain has $S_2 = n$ exactly. Smearing ($\epsilon = 0.1$) removes this spurious coherence of near-chains (their variance rises to 2.1 times the sprinkled value) and also tames the large fluctuations of the unsmeared 4D action on sprinklings (exponent 2.27 at $\epsilon = 1$, 1.84 at $\epsilon = 0.03$).

Variance alone cannot decide suppression. In the Gaussian approximation the coherent part of a class amplitude is $W\, e^{-\beta^2 \mathrm{Var}/2}$; once this falls below the random floor $\sqrt{W}$, the amplitude is $|A_\Gamma|^2 = E_\Gamma W$. A class of entropy W can therefore be pushed only to $\sqrt{W}$ by large variance, and for KR orders $\sqrt{W} = 2^{N^2/8}$ still dwarfs the $e^{O(N \log N)}$ entropy of manifoldlike classes. Beating entropy requires structural cancellation, $E_\Gamma \ll 1$.

C The KR threshold as a lock window

KR orders supply the cleanest example of structural cancellation. With $\epsilon = 1$ their 4D action is $S_4 \approx (4/\sqrt{6})(N - N_0)$, where the link count N_0 is binomial with $M = N^2/4$ trials and $p = 1/2$ (measured $\mathrm{Var}(S_4)/N^2 = 0.164$ against the predicted $(16/6)\,\mathrm{Var}(N_0)/N^2 = (8/3)(1/16) = 1/6$). The sum over link configurations is then an exact product,

$$|A_{\mathrm{KR}}| = \left|2\cos(\beta'/2)\right|^M, \qquad \beta' = \frac{4}{\sqrt{6}}\,\beta,$$

which falls below the amplitude of a single configuration iff $\beta' \in (2\pi/3, 4\pi/3)$, i.e. $\beta > 1.2825$. With $\beta = (\ell/\ell_p)^2$ the threshold is $\ell > 1.1325\,\ell_p$, within 0.3% of the bound $\ell \gtrsim 1.136\,\ell_p$ obtained by Carlip, Carlip and Surya for $d = 4$ [4, 10]. The suppression is a window, repeating every 2π in β', rather than a monotone effect. The mechanism is phase locking: every "site" (a potential link) shifts the action by the same amount.

D The defect-gas theorem

The non-manifold structure of Sec. VI is not of this kind. It consists of local defects of several types (twins, jagged clones, mutated clones, cut links) whose action shifts are widely spread. We model it as a defect gas.

Theorem 21 (defect gas). Let a background C_0 carry M sites, each with options $j \in \{0, 1, \dots, K\}$ ($j = 0$ clean), such that every combination is a distinct causal set and the action is additive, $S = S_0 + \sum_s \delta(s, j_s)$. Then

$$\sum_{\text{configurations}} e^{i\beta S} = e^{i\beta S_0} \prod_{s=1}^{M} P(s, \beta), \qquad P(s, \beta) = \sum_{j=0}^{K} e^{i\beta\delta(s,j)},$$

so the growth rate relative to the clean configuration is exactly $r_M(\beta) = M^{-1} \sum_s \ln \left| 1 + \sum_{j\geq 1} e^{i\theta_{s,j}} \right|$ with $\theta_{s,j} = \beta(\delta(s, j) - \delta(s, 0))$. If the phase vectors $\theta_s \in \mathbb{T}^K$ are pairwise independent and uniformly distributed, then almost surely the sum is nonzero for every M and

$$r_M(\beta) \longrightarrow m(1 + x_1 + \cdots + x_K) = \int_{\mathbb{T}^K} \ln \left| 1 + e^{i\theta_1} + \cdots + e^{i\theta_K} \right| d\theta,$$

the Mahler measure [31]. Moreover $m(1 + x) = 0$ and $m(1 + x_1 + \cdots + x_K) > 0$ for $K \geq 2$.

Proof. (i) The factorisation is the finite identity "sum of products equals product of sums" together with $e^{\sum} = \prod e$; the common factor $e^{i\beta\delta(s,0)}$ has modulus one. (ii) The rate is the empirical mean of $\ln |W_K(\theta_s)|$ with $W_K(\theta) = 1 + \sum_j e^{i\theta_j}$. Integrating out one phase with Jensen's formula, $\int \ln |e^{it} + w|\, dt/2\pi = \log^+ |w|$, and using the uniform bound $\int |\ln |e^{it} + w||\, dt/2\pi \leq 2 \ln(1 + |w|)$ with Tonelli's theorem, $\ln |W_K|$ is integrable and the strong law of large numbers gives the limit. (iii) The same inner Jensen step gives $m(1 + x_1 + \cdots + x_K) = \mathbb{E} \log^+ |1 + x_2 + \cdots + x_K|$, which is $\log^+ 1 = 0$ for $K = 1$. For $K \geq 2$ it is positive, because on the box $\theta_j \in (0, \pi/2)$, of measure $4^{-(K-1)}$, the real part of $1 + x_2 + \cdots + x_K$ exceeds 1. (iv) On each slice, $e^{it} = c$ has at most one solution in $(0, 2\pi]$, so the zero set of W_K is null. □

The values are $m(1 + x + y) = (3\sqrt{3}/4\pi) L(\chi_{-3}, 2) = 0.3231$ and $m(1 + x + y + z) = 7\zeta(3)/2\pi^2 = 0.4263$ [32], and numerically $m = 0.5444$ for $K = 4$; for large K, $m \approx \frac{1}{2} \ln K$. The classical entropy per site is $\ln(K + 1)$, so interference removes roughly half of it, the analytic form of the random floor of Sec. VII B. Suppression below the clean configuration ($r < 0$) requires phases far from uniform. For a single shift δ, $|1 + e^{i\beta\delta}| < 1$ iff $\cos \beta\delta < -1/2$: all sites must sit in the lock window simultaneously, which is the KR situation of Sec. VII C. Every step of the proof, including the integrability, the Fubini step, the null zero set and the strong-law substitution, is checked in Lean (Sec. VIII); uniformity of the phases is an explicit hypothesis there.

E Tests of the hypotheses

Shifts and additivity. On 2D sprinkled diamonds ($N = 300$, every site) the single-defect shifts are broad: for the unsmeared action their standard deviation is 25–44 (cuts smallest), for $\epsilon = 0.1$ it is 0.06–0.20. In 4D they are broader still. Five far-apart twins shift the action by -370 against a sum of singles of -366; five far-apart cuts give -40 against -40.

Uniformity of the phases. For integer-valued shifts the phases are periodic in β (period π for the unsmeared 2D action, whose shifts are even), so we scan one period and compare the empirical rate with the Mahler value.

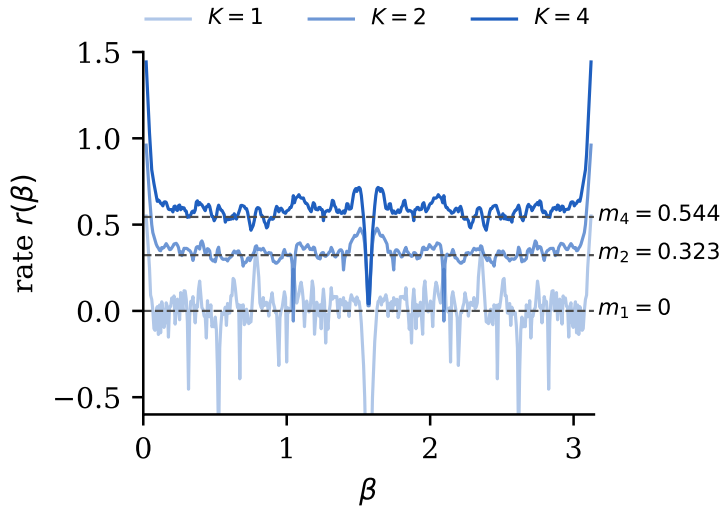


Figure 2: Empirical defect-gas rate $r(\beta) = M^{-1} \sum_s \ln |1 + \sum_{j \le K} e^{i\beta\delta_{s,j}}|$ over $M = 300$ sites of a 2D sprinkled diamond (unsmeared action), for one defect type ($K = 1$, twins), two ($K = 2$, twins and cuts) and four. Dashed lines: Mahler measures. Dips mark lattice resonances of the integer-valued action.

Setting	Phase non-uniformity, median over β (uniform ≈ 0.058)	$K = 2$: rate at generic β (Mahler 0.323)	$K = 2$: fraction of β with $r < 0$	$K = 4$: minimum rate over β (Mahler 0.544)
2D, $\epsilon = 1$	0.050–0.066	0.336 ± 0.045	0.5%	+0.042
2D, $\epsilon = 0.1$	0.55–0.94	0.318 ± 0.041	0	+0.459
4D, $\epsilon = 1$	0.048–0.058; cuts 0.46	(cuts never generic)	0.5%	+0.239
4D, $\epsilon = 0.1$	0.11–0.93	(never generic)	0	+0.346

Non-uniformity is the median over β of $|\langle e^{i\beta\delta} \rangle|$ across the four defect types; "generic" β are those where every involved type is within three times the uniform level. For the unsmeared 2D action the phases are as uniform as 300 sites allow at 88% of β, and the rate there matches Mahler for $K = 2$ (0.336 against 0.323) and $K = 4$ (0.585 against 0.544). For one type the β-average is +0.014, consistent with Jensen's zero, with sign changes across β. Where uniformity fails, as for small smeared shifts at moderate β or for 4D cuts, the phases bunch near zero and the rate rises above the Mahler value. Suppression ($r < 0$) occurs only in narrow lattice resonances, on 0.5% of β for $K = 2$ and never for $K = 4$.

Overlap and interaction. Overlapping or causally related defects are not additive. We built every configuration explicitly (cuts first, then twins copying the cut relations, twin–twin relations following their sites) on $N = 200$ sprinkled diamonds and summed the phases exactly, for dilute sites (farthest-point sampling) and dense sites (nearest to the centre), averaged over three backgrounds.

Sites, defect types	$M = 4$	$M = 8$	$M = 12$
Dense, $K = 1$ (twins)	0.214 ± 0.042 / 0.006	0.265 ± 0.011 / 0.004	0.306 ± 0.006 / 0.001

Sites, defect types	$M = 4$	$M = 8$	$M = 12$
Dilute, $K = 1$	0.076 ± 0.059 / 0.015	0.127 ± 0.056 / 0.008	0.137 ± 0.032 / 0.006
Dense, $K = 2$ (twins, cuts)	0.437 ± 0.020 / 0.327	0.477 ± 0.022 / 0.358	0.515 ± 0.006 / 0.346
Dilute, $K = 2$	0.411 ± 0.048 / 0.385	0.383 ± 0.029 / 0.317	0.424 ± 0.036 / 0.340

Each cell gives the β-averaged rate from the exact configuration sum (mean $\pm$ standard error over three backgrounds) and, after the slash, the product formula of Theorem 21 built from the single-defect shifts. The exact rate exceeds the product formula at every M and in every setting, and the excess does not shrink as M grows. For dense twins the exact rate reaches 0.317 ± 0.006 at $M = 16$ while the product (Jensen) value stays at zero; for dense two-type defects the excess is 0.10–0.17. At individual β the exact rate is higher on 66–92% of the scan, and the fraction of β with a negative exact rate falls with M (dense, $K = 2$: from 2.9% at $M = 4$ to zero for $M \geq 10$). Because the background is fixed ($N = 200$), larger M also means more closely spaced sites, so the M-dependence mixes system size with defect density. The robust conclusion is that interactions act like additional phase randomness and raise the rate.

No double counting. The path sum counts each unlabelled causal set once, so distinct configurations must give non-isomorphic posets. For $M = 8$, $K = 2$, two backgrounds and both site sets, all $3^8 = 6561$ configurations receive distinct Weisfeiler–Lehman invariants (refinement over past and future colour multisets). Distinct invariants imply non-isomorphism, so no configuration is counted twice. At $N = 8$, where every poset can be enumerated, the non-2D orders produced by the classical rules sit at the random level, $E = 0.86$–1.23, as the theorem predicts.

F Conclusion of the quantum test

Interference on a classical growth tree does not remove the multi-type local defects generated by order-only rules: their amplitude grows at the Mahler rate, about half the classical entropy rate, and interactions between defects raise it further. Interference does remove classes whose action shifts phase-lock, and the KR threshold is the clean instance. Whether a class is suppressed is thus a computable property of its shift distribution, not of its variance.

VIII Formal verification in Lean 4

We used Lean 4 [33] with Mathlib [34] as a check on the proofs, not as a substitute for them. Five files compile without `sorry`; every theorem depends only on the standard axioms `propext`, `Classical.choice` and `Quot.sound`. The toolchain is Lean 4.36.0-rc1 with Mathlib built from source; all declarations live in the namespace `CausalDensification`.

File	What is verified	Theorems
`UndampedKernelConstants.lean`	Theorem 18 constants: bounds on $Z(t)$, $\int_0^T \lvert x - t\rvert \, dx = Z(t)$, stationarity $\int Z(t)q(x\lvert t)\, dt = Z(x)$, $\int Z = 1/24$, $T/Z \leq 8$; Lipschitz constant 80 of q; bounds on the shell-overlap length ℓ; $h_{abc} \geq 0.035$; the Doeblin constant; the L2 constants 288, 20 and 436	19

File	What is verified	Theorems
`PosetLimitCore.lean`	Theorem 17 for pairs on $\mathbb{R} \times \mathbb{R}^3$ (product-measure continuity, portmanteau, boundary of the causal relation contained in null pairs); summability for Theorem 19; for Theorem 20 the Pólya identities, the exact same-class probability $2(N-M)/((M+1)(N-1)) \le 2/(M+1)$, the shell volume bound, the Jacobian cancellation and the choice $M = \lceil \varepsilon^{-1/2} \rceil$	14
`PosetLimitFramework.lean`	Theorem 17 for general k and on slab $\times \mathbb{T}^3$ with minimum-image distance; the Fatou step of Theorem 20; summable second moments imply a.s. convergence; Doeblin implies uniform geometric ergodicity; bounded martingale differences plus summable drift imply a.s. convergence; the telescoping identity of Theorem 18	28
`DefectGasAlgebra.lean`	Theorem 21: exact factorisation; Jensen's formula for $1+x$; the inner Jensen identity; the lock window	4
`DefectGasLimit.lean`	Theorem 21: integrability and the Fubini step on the torus, $m(1+x_1+\cdots+x_K) = \int_{\mathbb{T}^{K-1}} \log^+ \lvert 1+x_2+\cdots+x_K \rvert$ (the Lean statement indexes the K phases from 0); $m = 0$ for $K = 1$ and $m > 0$ for $K \ge 2$; the zero set is null; the strong-law substitution; the full statement of Theorem 21	24

The checks found two errors, both in bookkeeping rather than in mathematics. A Pólya same-class probability had been written with an $O(1/N)$ correction that does not exist; the exact finite-N formula has none. And a summation index in the L^1 bound of Theorem 19 had been typed so that Lean read it as a real number, for which the series is defined to be zero; with a natural-number index the proof goes through.

Two layers remain unformalised. The first consists of analysis tools missing from Mathlib: the three-shell convolution formula and the $W^{1,1}$ translation estimate $\mathrm{TV}(\mu, \mu_e) \le \frac{1}{2}|e|\,\|\nabla p\|_1$. The second is modelling: writing the replication process as a stochastic process in Lean and verifying the hypotheses of the abstract theorems (the minorisation of K^5, stationarity, the perturbation bound L3, autonomy of the time process). In Theorem 21 the uniform-phase hypothesis is an explicit assumption of the Lean statement, not a theorem.

IX Discussion and open problems

The replication model answers the four problems in different currencies. The continuum limit is proved: for every damping the process has a poset limit, a uniform sprinkling of a conformally flat spacetime, and below the damping length the grown set satisfies high-probability, uniform versions of Madsen's F2 and F3 for the conformal metric. Lorentz invariance at the discreteness scale is established numerically to a few per cent, with violation confined to the clumping excess. The rogue-poset problem is answered by verification rather than by an order-intrinsic dynamics: the

model's output passes every intrinsic test, and three families of order-only rules fail for a single geometric reason. On quantum dynamics the result is negative but precise. Phase interference suppresses a class only when its action shifts phase-lock, as for layered and KR orders. Multi-type local defects grow at the Mahler rate, and interactions between overlapping defects make them grow faster.

The negative result fits the positive ones. Section VI shows that order-only local rules generate jagged, multi-type defects; Section VII shows that $e^{i\beta S_{\mathrm{BD}}}$ cannot remove them. A viable dynamics must therefore carry coherent light-cone structure from the start, as the coordinate model does. Interference is still useful: it removes the entropically dominant layered orders that no classical rule would want to produce in the first place.

Several problems remain open.

1. **An order-intrinsic 4D dynamics.** The most promising route we see uses the link-level mutation of Sec. VI B as a proposal and the local BD action or the abundance spectrum as a Metropolis acceptance, letting the action select coherent light cones.
2. **The uniform-phase hypothesis.** For the unsmeared action the shifts are integers, so uniformity at generic β should follow from an equidistribution argument once the shift distribution is controlled. Smeared actions need a separate treatment, although there the failure of uniformity only raises the rate.
3. **The large-scale conformal factor.** Replication fixes it by boundary conditions. Whether the Lorentzian path sum selects an expanding FRW profile on replicated sets is untested.
4. **Sharper constants.** Theorem 8 is about 2.7 times looser than the measured supremum; the F3 bounds stop at the Talagrand $\sqrt{L}$ scale, while true fluctuations are smaller [23, 35]; and the law of the random limit profile for $\alpha > 0$ is unknown.
5. **The Lean instantiation layer** (Sec. VIII).

All simulations use the scripts of the accompanying research log (Python with NumPy; a C program for chain heights), and the five Lean files are available with it.

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